



3D Scanning Lab



ALMA MATER STUDIORUM
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DIPARTIMENTO
DI MATEMATICA

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Hardware Components Overview

- **Key Elements in a 3D Scanner:**

- **Optical Sensor:** Captures depth information (laser, structured light, or time-of-flight).

- The choice of hardware impacts resolution, speed, and accuracy.



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 - **Camera(s):** Acquire color (RGB) data, aiding in texturing.
 - **Processing Unit:** Onboard or external computation for real-time point cloud generation.
- The choice of hardware impacts resolution, speed, and accuracy.



Typical Workflow of 3D Scanning

- **Preparation:** Object placement and environment setup (lighting, background, exposure)
- **Acquisition:** Sensor scanning and collection of raw point cloud data
- **Registration:** Align multiple scans
- **Postprocessing:** Noise filtering, mesh reconstruction, hole filling, and texture mapping
- **Visualization and Export:** Final 3D model for further analysis or rendering



Challenges/Considerations

- **Accuracy vs. Speed:** Trade-offs in data acquisition and processing times.
- **Calibration:** Importance of proper calibration to minimize mathematical and systematic errors.
- **Environmental Factors:** Impact of lighting, reflectivity, and occlusions on scan quality
- **Postprocessing Complexity:** Balancing detail with computational requirements

Hands-on 3D Scanners

- Revopoint Mini 2



- Shining 3D Einstar



Hands-on 3D Scanners cont.

- Revopoint Mini 2



- Shining 3D Einstar



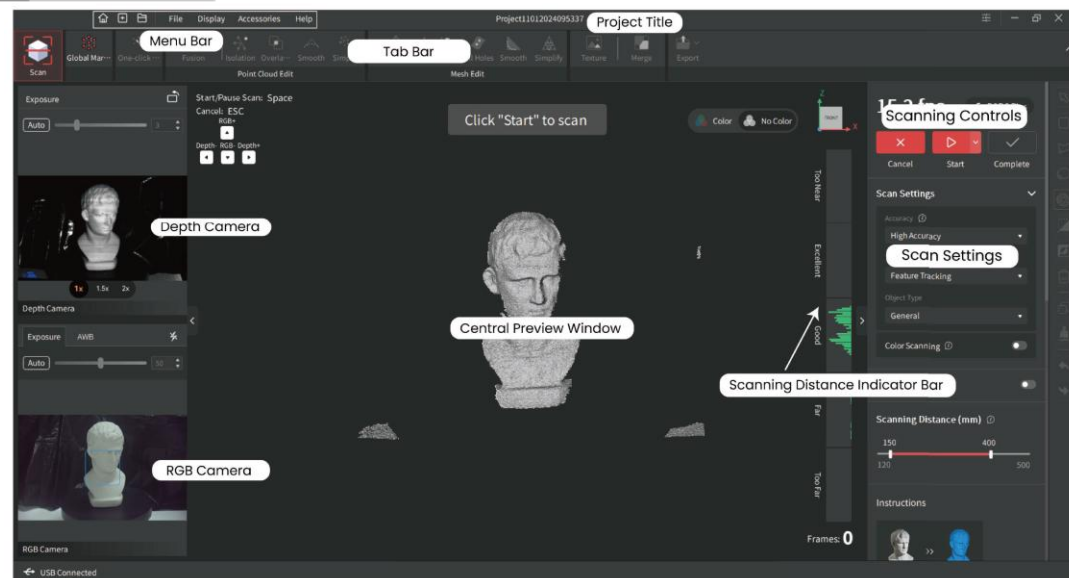
Construction	Tabletop / Handheld	Handheld
Technology	Dual-camera Blue Structured Light	Infrared VCSEL structured light
Working Distance	120 - 250 mm	160 mm-1400 mm, opt. 400 mm
Point Distance	Up to 0.02 mm	0.1 mm – 3 mm
Field of view	168 x 132 mm at 250 mm	434mm x 379mm at 400mm
Scannable objects	100 x 100 x 100 mm	200 x 200 x 200 mm
Texture scan	yes	yes
Software	RevoScan	EXStar in “Guest Mode”
Special requirements		NVIDIA graphics card

RevoPoint 5: <https://global.revopoint3d.com/en-eur/pages/support-download>

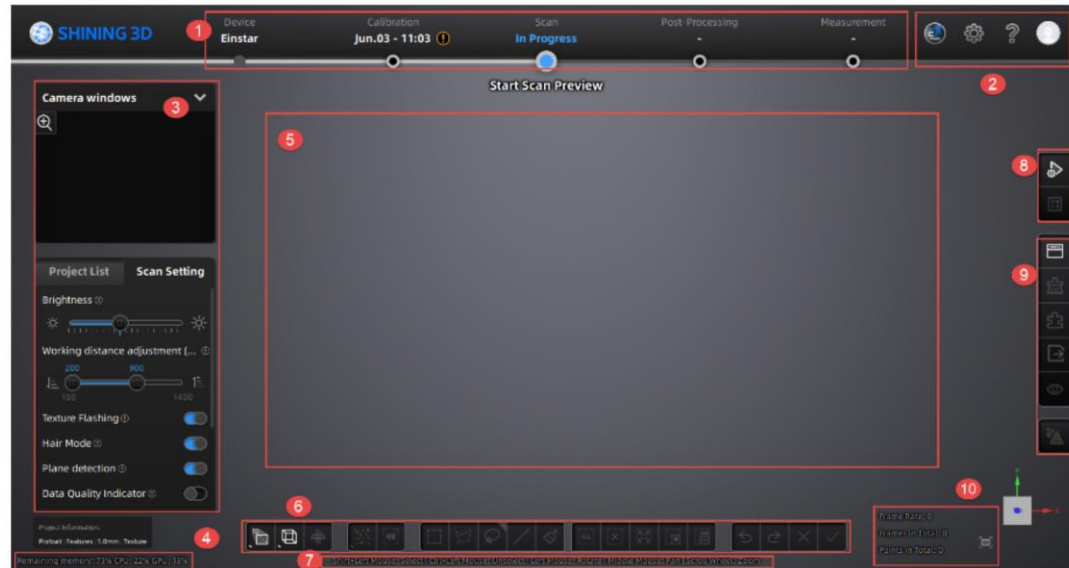
EXStar: <https://support.einstar.com/support/solutions/articles/60000782576-einstar-software-download>

Scanning Interface

Revo Scan



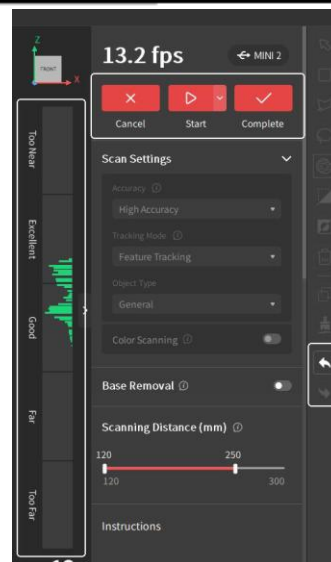
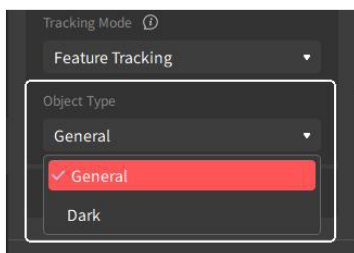
Shining EXStar



Preparation: Scanning Settings

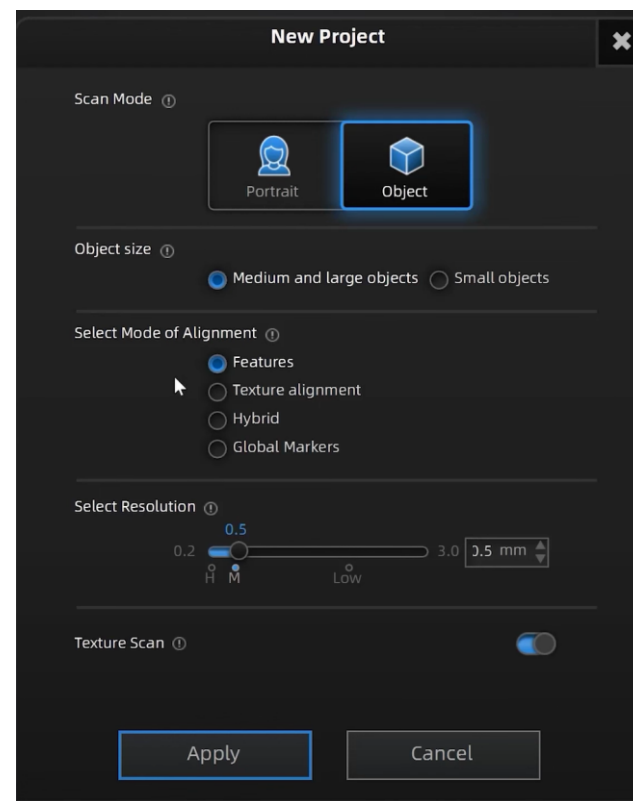
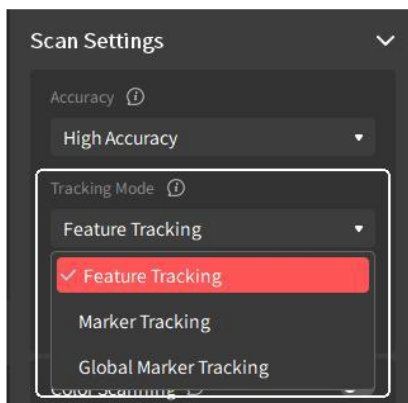
- **Object type**

- Define physical properties of your object to be scanned



- **Tracking: Feature/Marker/Hybrid**

- Define if scanning markers will be used



Preparation: Scanning Settings cont.

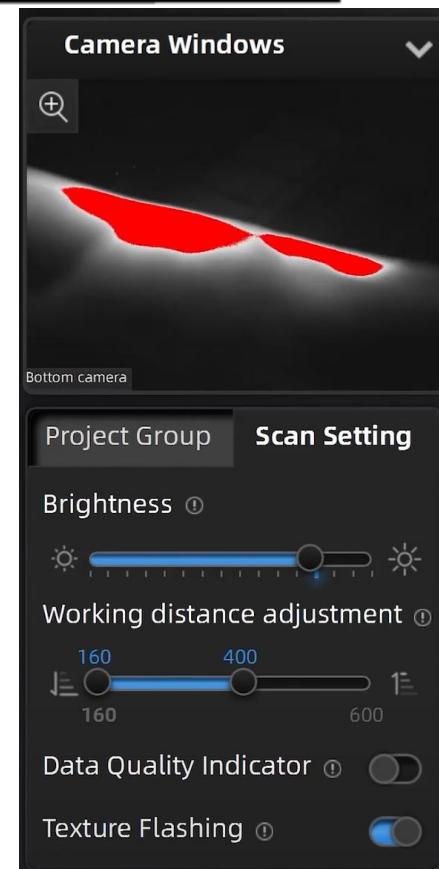
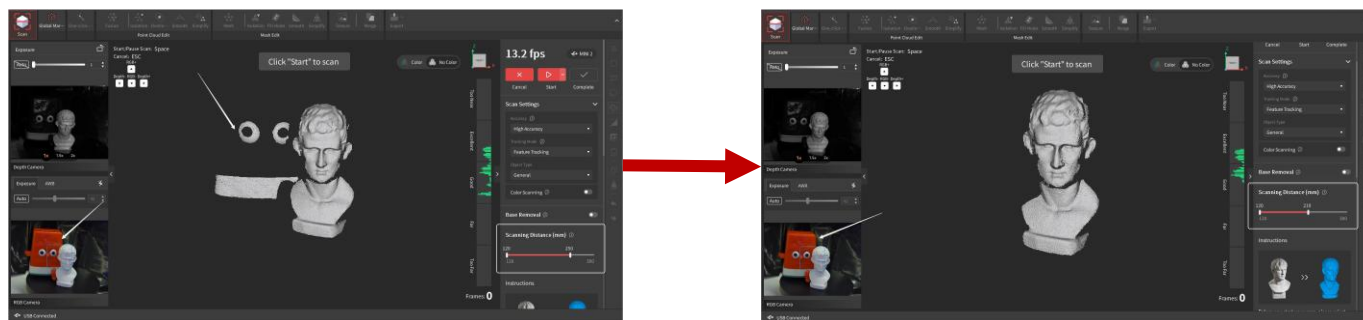
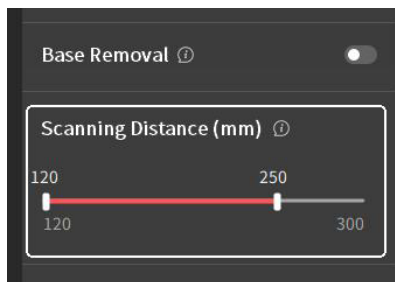
• Exposure

- Correct white balance of the cameras



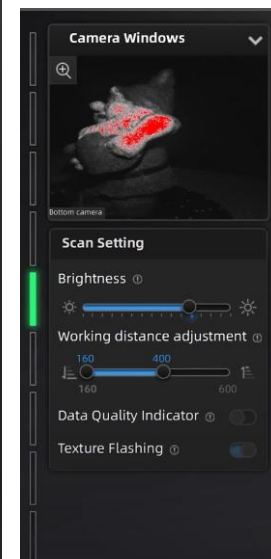
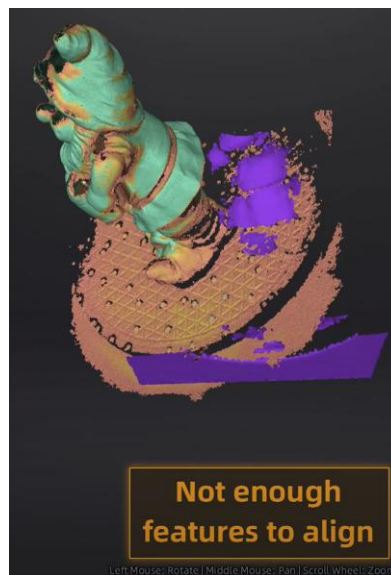
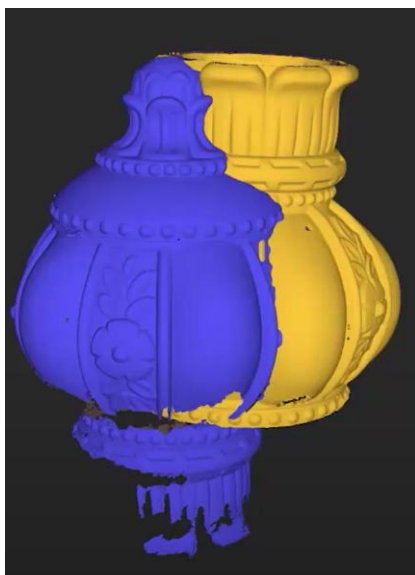
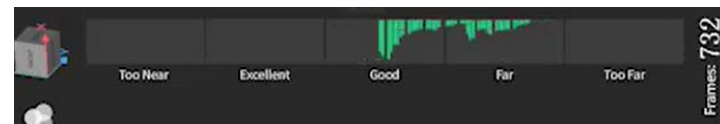
• Scanning Distance

- Tune scanning range



Scanning Process

- Mind the distance to the object
- Use of quality map
- Case-specific approach for hard to scan objects
- Pause – Check – Adjust
- Possibility of multiple scans

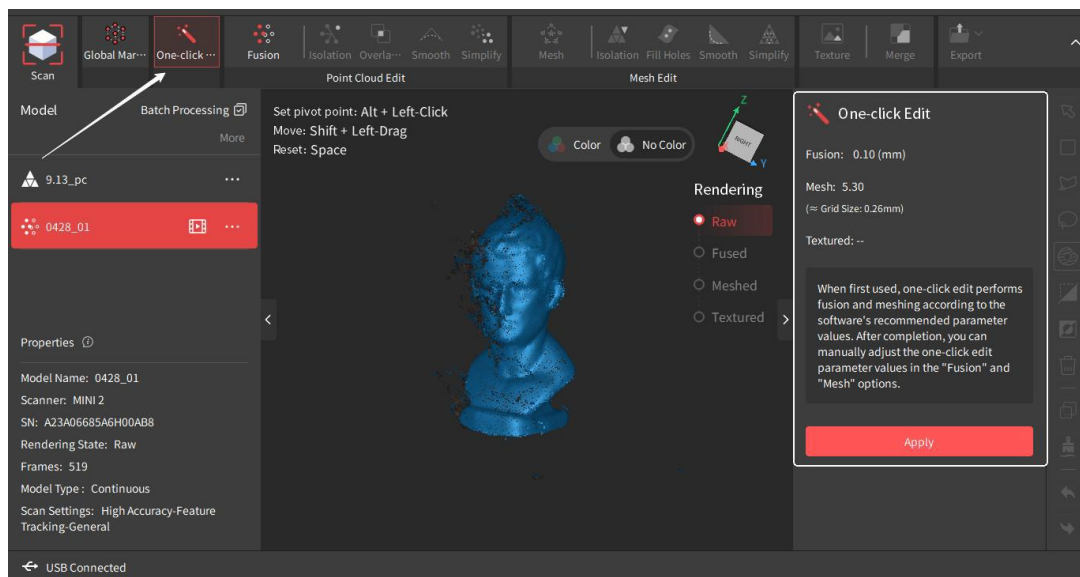


Postprocessing/Editing of 3D Scans

- Can be applied in part directly to the point cloud or to the generated mesh:
 - **Data Cleaning:** Removing noise and outliers from point clouds.
 - **Registration and Alignment:** Fine-tuning alignment of single frames
 - **Mesh Generation:** Converting point clouds to polygon meshes (Poisson reconstruction, Marching cubes).
 - **Hole Filling:** Enclosing areas with missing data
 - **Texture Mapping:** Applying color/texture to mesh vertices
 - **Optimization:** Reducing model complexity for real-time applications (e.g., gaming, VR).
- Many tasks are automated for smoother user experience
- Various software available: RevoScan, EXStar, Blender, MeshLab, etc.

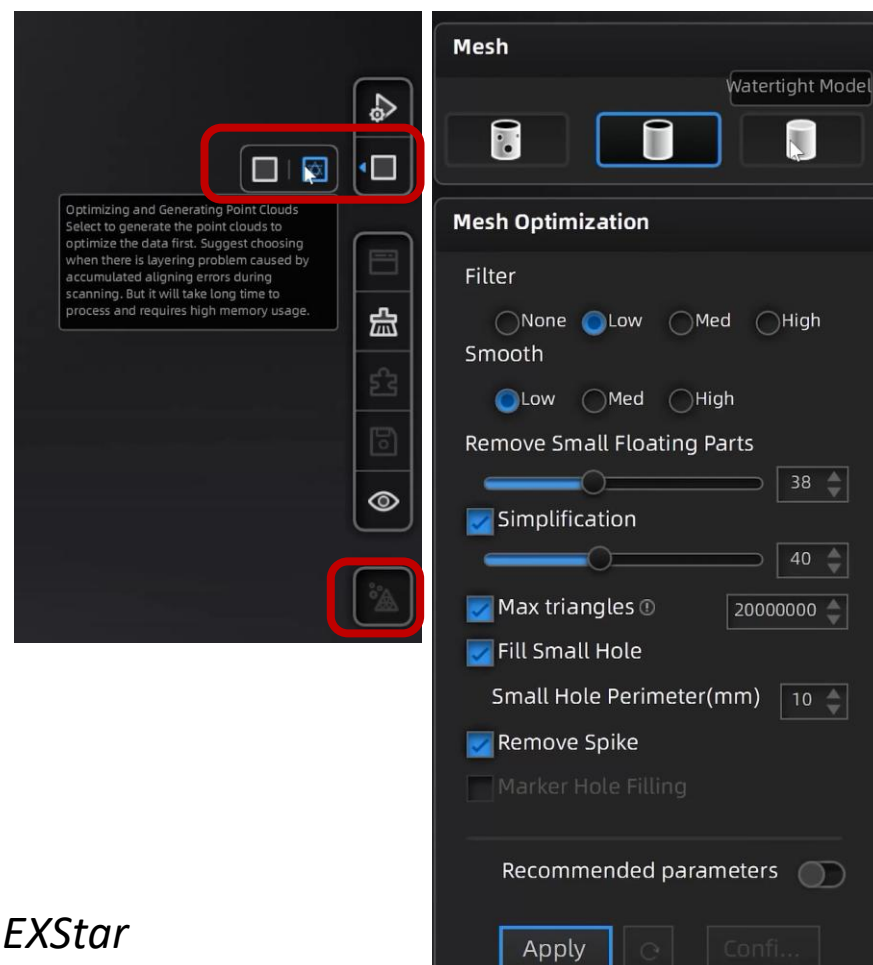
Mesh Generation

1. One-click Edit
or
1. Point-Cloud Fusion and Processing
2. Mesh generation



RevoScan

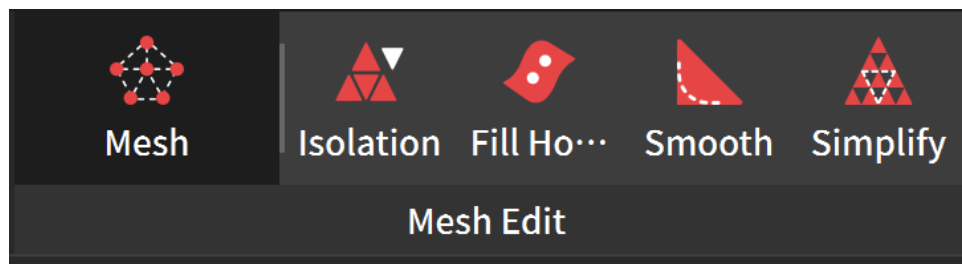
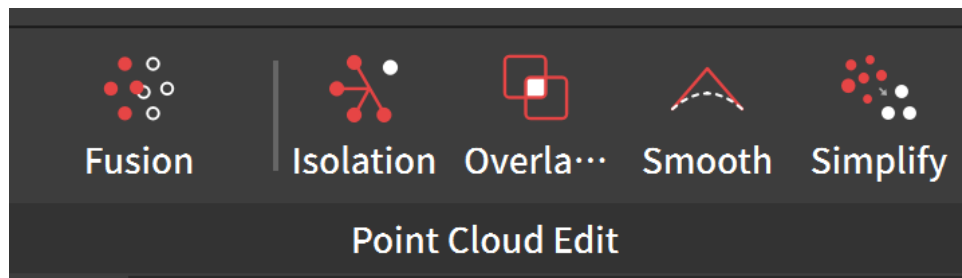
1. Point-Cloud generation
2. Mesh generation



EXStar

Post-Processing Functions

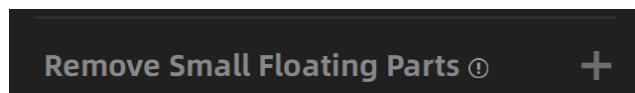
- Eliminate isolated parts
- Smooth the object
- Fill mesh holes
- Simplify triangulation



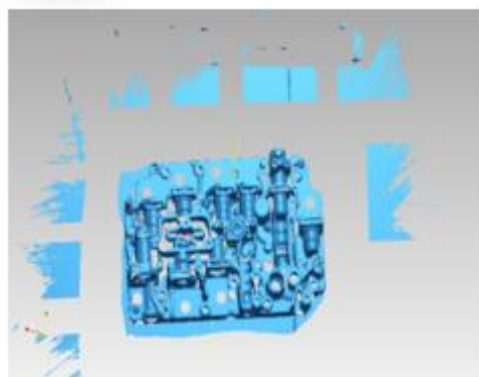
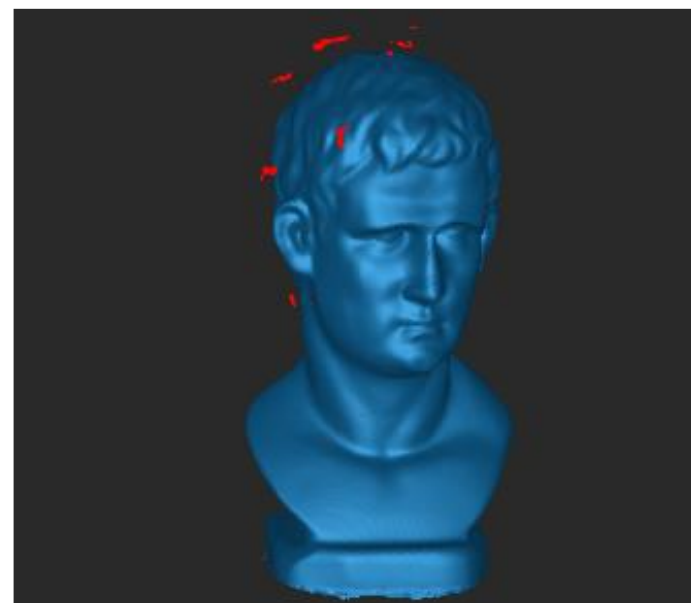
Mesh Editing	
Texture Adjustment ⓘ	+
Simplification ⓘ	+
Mesh Optimization ⓘ	+
Smooth ⓘ	+
Remove Small Floating Parts ⓘ	+
Auto Hole Filling ⓘ	+
Manual Hole Filling ⓘ	+
Flip Normal ⓘ	+
Cutting Plane Tool ⓘ	+
Mirror ⓘ	+
Zoom ⓘ	+

Post-Processing Functions cont.

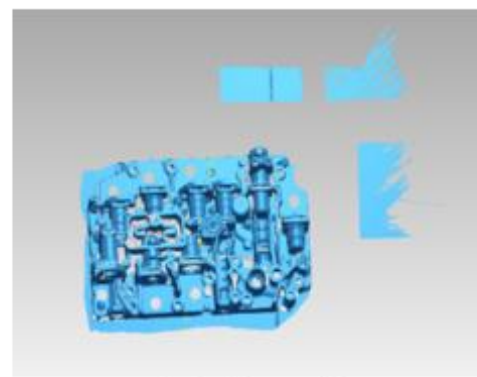
- Eliminate isolated parts



- Selects and removes connected components that are not connected to the data mesh



Original data



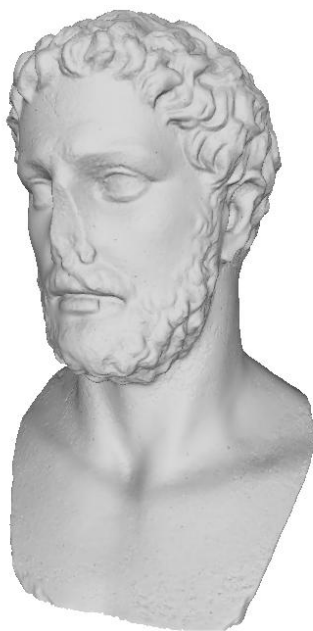
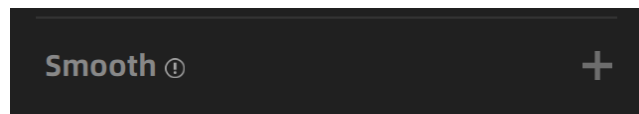
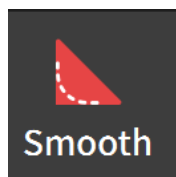
Set value as 50



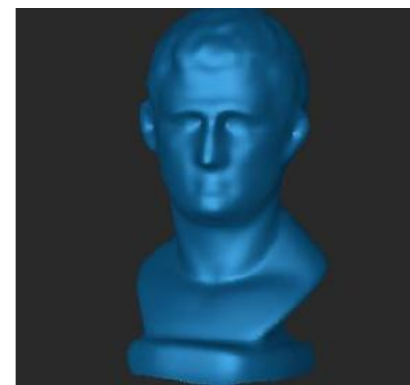
Set value as 100

Post-Processing Functions cont.

- Eliminate isolated parts
- Smooth the object



Before

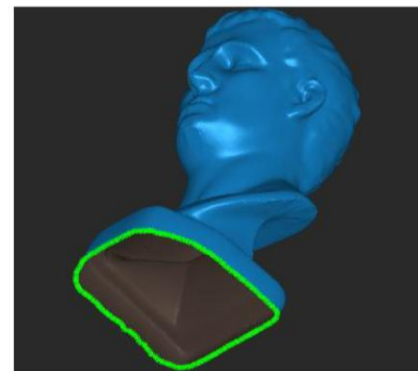
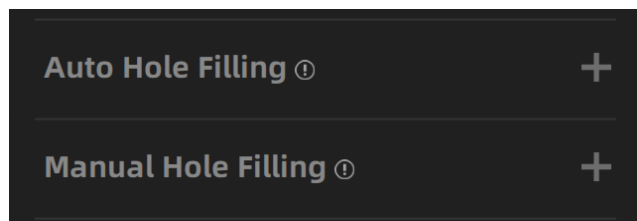
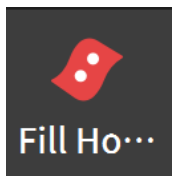


After

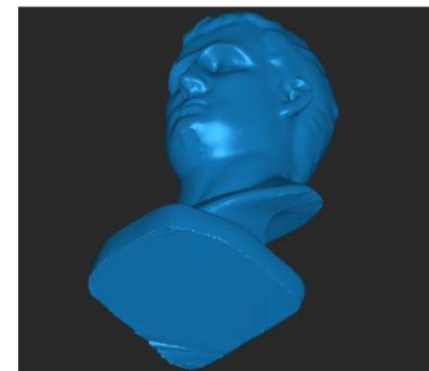
- Eliminates noise from noisy mesh
- Removes fine details

Post-Processing Functions cont.

- Eliminate isolated parts
- Smooth the object
- Fill mesh holes

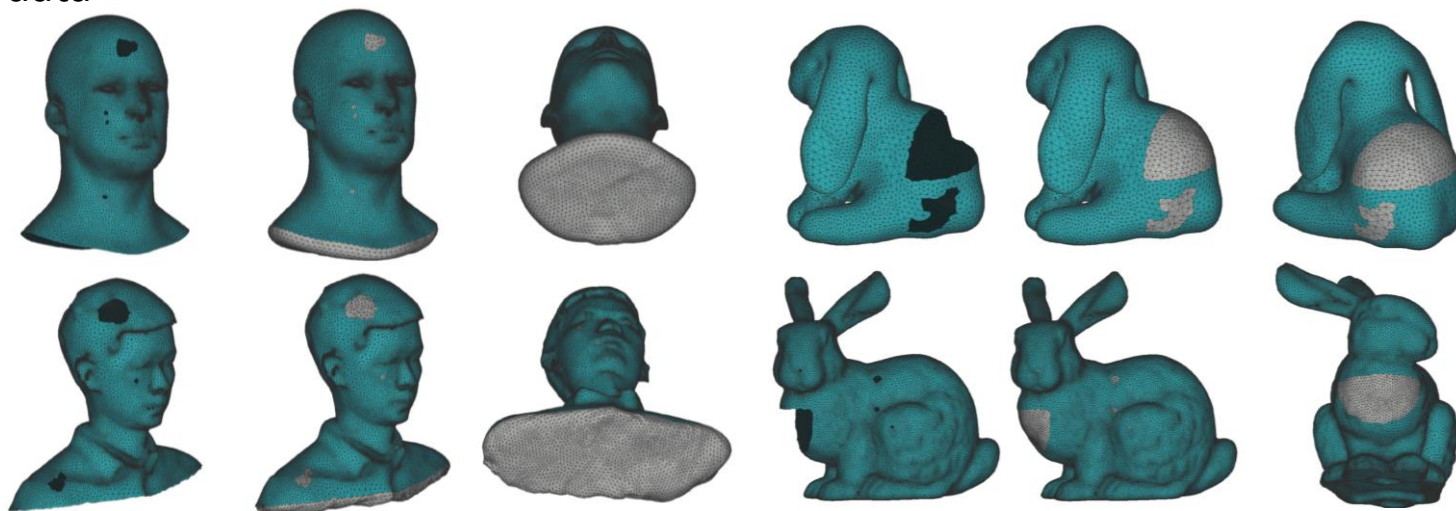


Before (Hole Filling)



After (Hole Filling)

- Encloses holes to create a watertight surface
- Fills in missing data



Post-Processing Functions cont.

- Eliminate isolated parts
- Smooth the object
- Fill mesh holes
- Simplify triangulation



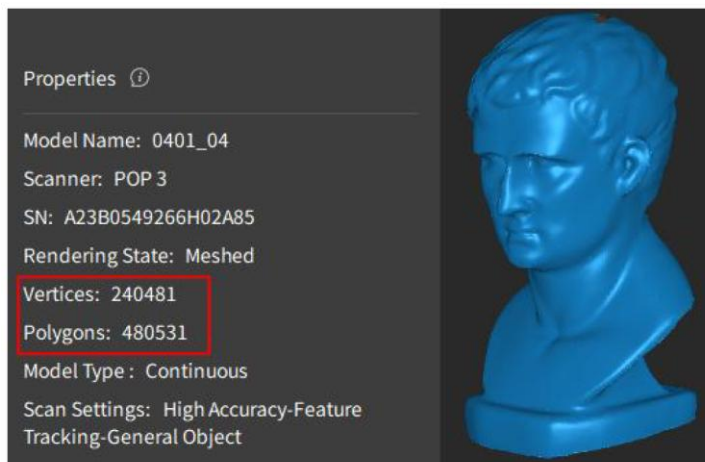
Simplification ⓘ



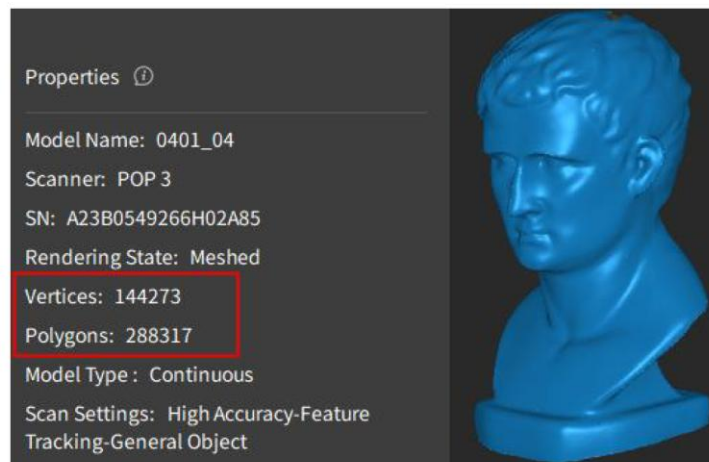
Mesh Optimization ⓘ



- Decreases the complexity – number of vertices and triangles of the mesh



Unsimplified



simplified



Useful links

- Mesh and project repository folder [3D scanning](#)
- Software [RevoScan](#), [EXStar](#)
- User Manual [RevoScan manual](#), [EXStar manual](#)
- Demo [RevoScan YT](#), [EXStar YT](#)